Closing the loop on aerospace alloys

Autonomous discovery via additive manufacturing

Sterling G. Baird

Vertical Cloud Lab · Mechanical Engineering, BYU

Utah AI Convergence 2026 · Advanced Manufacturing Panel

We import the elements our best alloys depend on

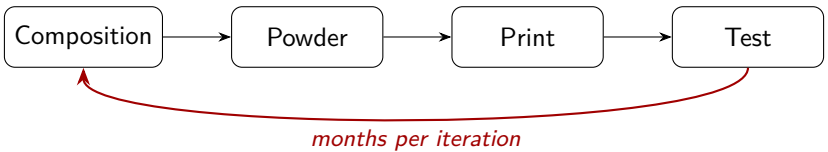
- **100%** of manganese — imported
- **100%** of zirconium concentrates — imported
- **74%** of chromium — imported

These are the workhorses of high-performance aluminum alloys for additive manufacturing.

The exposure

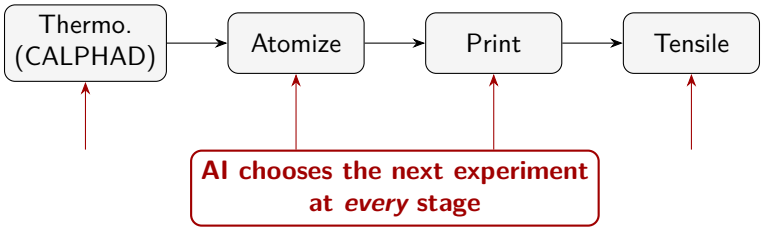
One supply disruption can idle a domestic AM production line.

The bottleneck is the development loop, not the printer



Each turn of the loop is slow, manual, and run *one alloy at a time*.

AI can drive the whole pipeline, stage by stage



Model each stage as a node; optimize the workflow end-to-end, cost-aware.

Flaime: same loop, run automatically overnight

Screen in silico, in parallel

- $\sim$23,000 element combinations
- thermodynamics + solidification, run in parallel

Validate the few that survive

- top 3–6 systems, 100–200 compositions
- powder $\rightarrow$ print $\rightarrow$ tensile

Phase I target (not yet measured)

Aim: $\sim\mathbf{10}\times$ faster powder-to-property
(planning estimate: $\sim 3\times$ smarter selection $\times$ $\sim 3\times$ closed-loop execution)

Phase I goal

Concept $\rightarrow$ lab-validated closed loop.

These tools transfer to any staged manufacturing workflow

Start optimization without custom code

HONEGUMI generates a runnable optimization template from a few toggles.

Use cheap screens before expensive tests

Stage-aware optimization kills bad candidates early.

Prototype automation on cheap stand-ins

Frugal twins de-risk capital purchases before you buy the expensive machine.

Make experiments shareable and reproducible

Cloud execution plus open, reusable (FAIR) data and models.

Run the loop like software: automate it, share it.

Sterling Baird · github.com/vertical-cloud-lab

Honegumi: honegumi.readthedocs.io · pip install honegumi